

EXAM 7 – SPRING 2015

7. (1.5 points)

An insurer writes Personal Automobile and Homeowners insurance in multiple geographic locations.

Identify three questions an actuary could ask company management to help determine how to segment the claims portfolio into appropriate classes for estimating unpaid claims liabilities. For each question, briefly explain why it should be asked.

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QUESTION 7	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE: A5: Describe the various sources of risk and uncertainty that are associated with the determination of reserves. Calculate risk margins that consider these sources of risk and uncertainty.
SAMPLE ANSWERS	
<i>Sample Answer 1</i> 1) Are we selling a wide range of policy limits? Asking this because different limited losses develop very differently and may want to group them into groups. 2) Are we writing a lot in CAT prone areas? Asking this because CAT vs nonCAT losses develop differently and may want to separate if have a lot of cat exposures. 3) Is there any expectation of legislative changes in some major states? This may have an impact on the auto liab. outstanding claims, e.g. if a court has been more pro-plaintiff, etc. <i>Sample Answer 2</i> 1) Has there been any catastrophe event in any geographic location? This question is important because cats have a different dev pattern than other "normal" losses, so we should model catas losses separated of the rest. 2) Are the coverages the same in all geographic regions? This question is important because if the coverage is different in between regions, the dev. patterns are likely to be different so the actuary should model only the policies with the same coverage altogether. 3) Are there any regions where the claims handlers are very understaffed or overstaffed? If there are difference between the number of claim handlers and the number of claims in different geographic areas, then the time to settle claims will be different and should be modeled for separately (or adjusted) <i>Sample Answer 3</i> 1) What are the coverages written under each line? Since different coverages have different development patterns, it is essential to group by coverage under each line. 2) What are the limits or deductibles used in underwriting? Since different limits/deductibles of policies have different development patterns. E.g. large limit may have a higher development later on. 3) Are there differences in regulation or other characteristics for the geographic locations? Since each location may have specific regulations, legislation, economic/social environment, the claim development patterns may be different. <i>Sample Answer 4</i> 1) Is homeowners exposed to catastrophe (event) risk? What/where are the events/locations	

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of concern?

Ask this question b/c we should separate catastrophes and non catastrophic claims for homeowners line due to different development patterns.

- 2) Are there different claims practices in different geographic regions?
If the company has 2 claims divisions, East and West, each w/ its own management, we should segment East vs. West auto and home claims b/c each region will have unique development patterns.
- 3) Are the coverages for personal auto unique, e.g. is there liability coverage and PD coverage? Are these handled by different departments?
Liability claims have a longer tail so it is appropriate to put these claims in their own class due to different development pattern than PD.

EXAMINERS' REPORT

The candidate was expected to know appropriate considerations for determining how to segment a portfolio for reserving analysis.

Common reasons for not receiving full credit included:

- Questions about data patterns – these would not be questions to ask of management but rather determined from looking at the data.
- Good questions, but weakly reasoned logic – it is not sufficient to simply say to group losses; need to know why it is important (e.g., different groups may have different development patterns).
- Questions about volume for credibility purposes – should be able to get that from data, not management.
- Questions only asking about deductibles, since for auto and homeowners the deductibles are relatively small and wouldn't materially impact development for segmentation.
- Questions about correlation between lines – more a consideration for risk margins rather than segmentation.